YAP 471 – Computational Finance Term Project Proposal

Project Title: Dynamic Sector and Asset Rotation with Markowitz Optimization

Team Members: Emirhan Yavuz, Aylin Barutçu, Bilgehan Tekin, Utku Kaya

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1. Asset Universe

- **Asset Class:** Equities and/or ETFs traded on Borsa Istanbul (BIST).
- **Selection Criteria:** We plan to select a universe of 5-10 liquid assets representing distinct 4-5 sectors or themes. We reserve the flexibility to include Exchange Traded Funds (ETFs) or Gold Certificates to better capture thematic exposures.
- **Tentative Asset Pool (Candidates):**
  i. **Mining/Gold :** Hedge against currency devaluation.
  ii. **Export Oriented :** Correlation with global demand FX rates.
  iii. **Interest Sensitive :** Sensitive to monetary policy cycles.
  iv. **Defensive/Retail :** Inflation pass-through capability.
- **Data Source:** Yahoo Finance ('yfinance') for daily prices; EVDS (CBRT) for macro indicators.
- **Sample Period:** Baseline of 3 years. If time and data processing constraints permit we plan to extend the backtest to a 5 or 10-year period, allowing us to capture multiple macroeconomic cycles. .

2. Primary Signal Direction

- **Chosen Signal Category:** Hybrid Approach (Price-based and/or Model-based).
- **Intended Signal Idea:** We aim to construct a **"Regime Score"** to adjust portfolio weights. This score will be derived from:
  - **Macro Factors:** Inflation (CPI), Policy Rates, USD/TRY changes.
  - **Technical Indicators:** Trend filters (Moving Averages), Momentum, or Volatility regimes.
- **Financial Intuition:** Assets perform differently under different economic and market conditions. By combining macro logic (e.g., "High Rates hurt REITs") with technical confirmation (e.g., "Price is below 200-day MA"), we aim to filter out false signals and improve risk-adjusted returns.

3. Initial Strategy Idea

- **Implementation Pipeline:**
  1. **Signal Generation:** Calculate a composite score (-1 to +1) for each asset based on the active Macro/Technical regime.
  2. **Dynamic Inputs:** Use these scores to tilt the *Expected Return Vector* (μ) or adjust the *Covariance Matrix* (Σ) fed into the optimizer.
  3. **Optimization:** Use the Markowitz Mean-Variance framework to find optimal weights subject to budget ($\sum w_i = 1$) and no-short constraints ($w_i \geq 0$).
- **Differentiation:** This strategy improves upon static diversification by actively reducing exposure to sectors that are technically weak or macro-economically disadvantaged.
- **Baseline Comparizon:** Equal-weight 1/N portfolio, rebalanced monthly.

4. Team Roles

- **Emirhan Yavuz:** Signal Generation (Coding the hybrid logic: Macro rules & Technical indicators).
- **Aylin Barutçu:** Data Pipeline (Fetching daily stock prices cleaning macro data).
- **Bilgehan Tekin:** Portfolio Construction (Implementing the Markowitz solver).
- **Utku Kaya:** Backtesting Reporting (Simulation, metrics, and visualization).

5. Expected Challenges

- **Signal Noise:** Technical indicators can generate false positives in choppy markets; combining them with macro data aims to filter this but adds complexity.
- **Look-ahead Bias:** Ensuring that macro data (often reported with a lag) is aligned correctly with trading dates.
- **Parameter Sensitivity:** The strategy's performance may depend heavily on the chosen look-back periods for indicators.